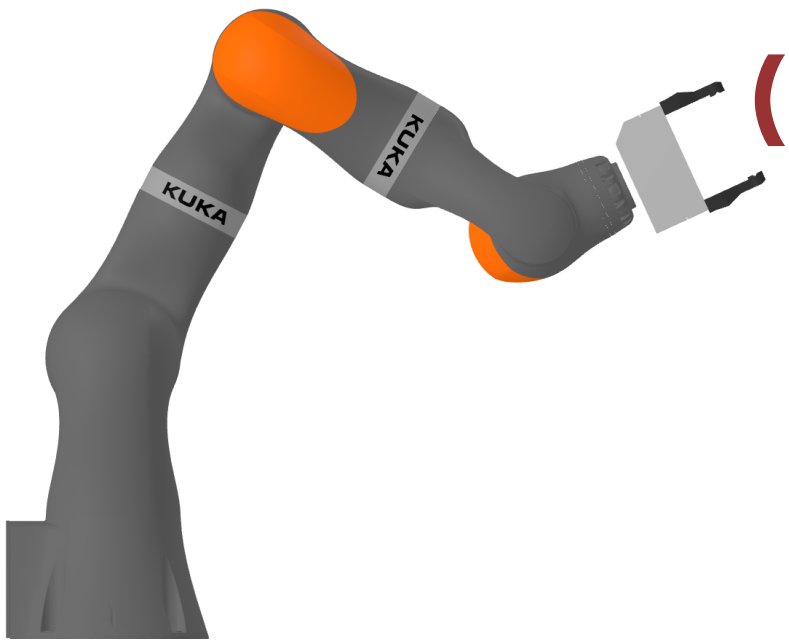


Deep Perception (for manipulation)

Part 1

MIT 6.4210/2

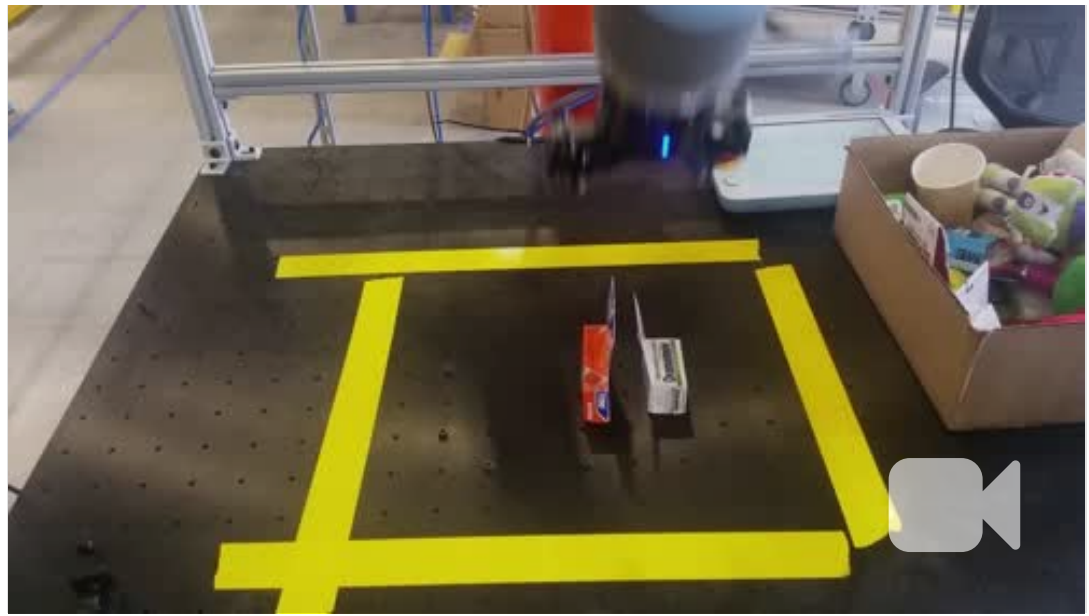
Robotic Manipulation

Fall 2025, Lecture 16

Massachusetts Institute of Technology

Limitations of using geometry only

- No understanding of what an object is.
  - "Double picks"
  - Might pick up a heavy object from one corner
- Partial views
- Depth returns don't work for transparent objects
- ...



- some tasks require object recognition! "pick the mustard bottles"

ImageNet: 14 Million labeled images

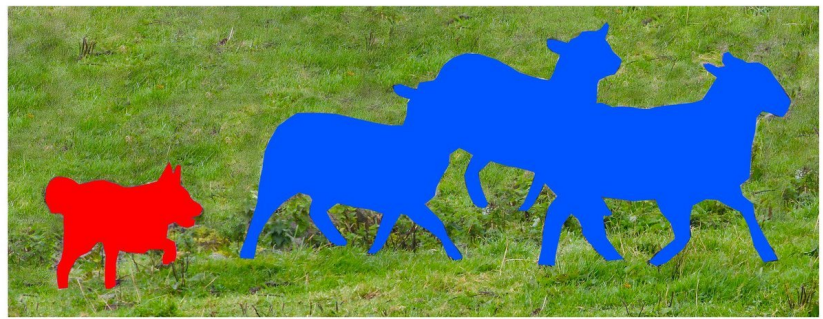


Released in 2009

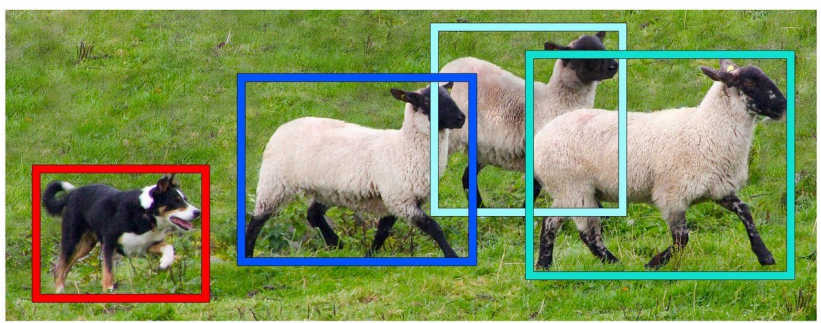
Traditional computer vision tasks



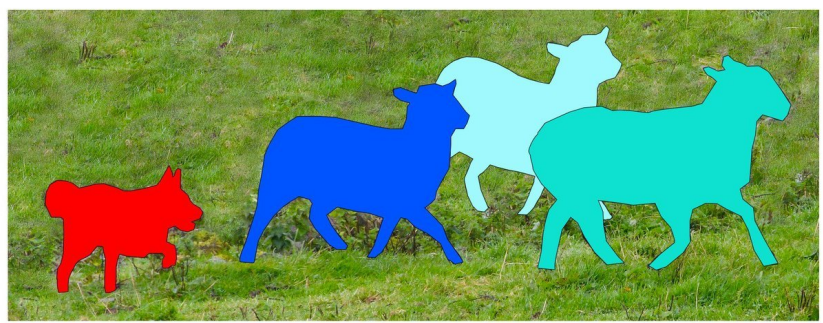
Image Recognition



Semantic Segmentation



Object Detection



Instance Segmentation

A sample annotated image from the COCO dataset

What object categories/labels are in COCO?

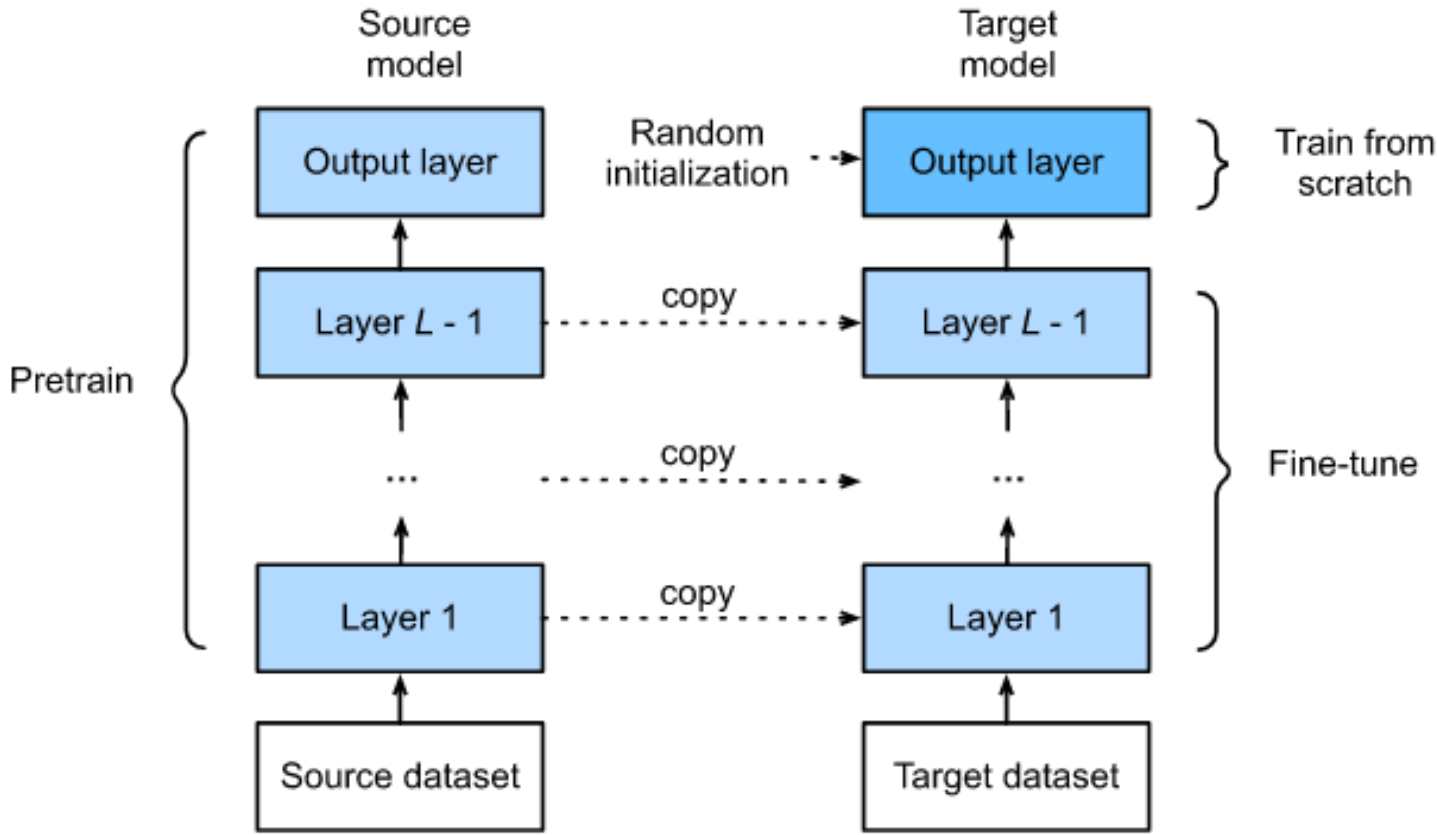
https://tech.amikelive.com/node-718/what-object-categories-labels-are-in-coco-dataset/#tablepress-2

Transfer learning

Something very powerful...

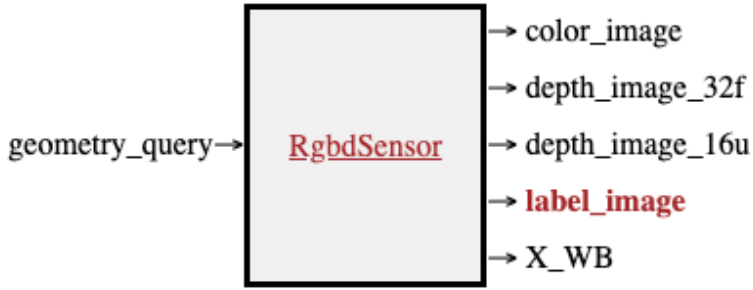
(Pre-)Training on ImageNet/COCO makes it easier to "learn" to recognize other objects

Fine tuning



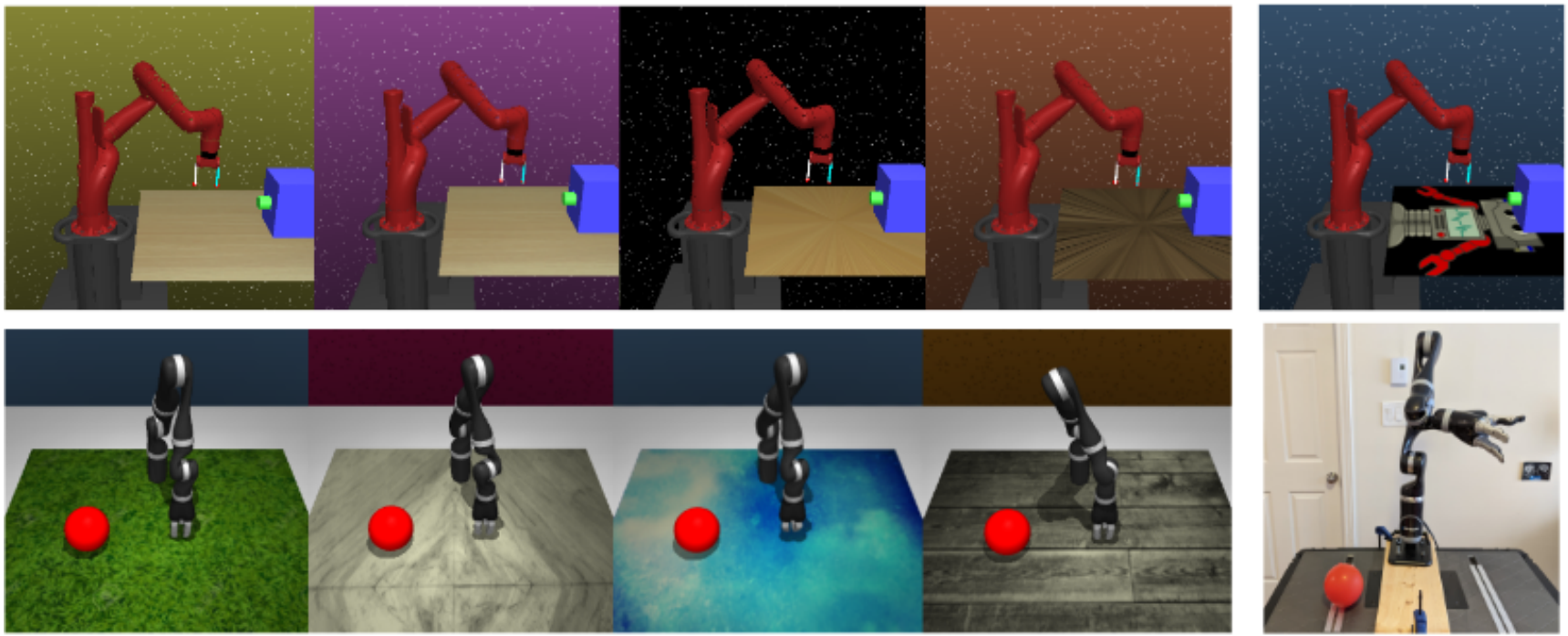
source: https://d2l.ai/chapter_computer-vision/fine-tuning.html

https://www.youtube.com/embed/nc65CF1W9B0?enablejsapi=1&mute=0&controls=0&rel=0&start=65


geometry_query
RgbdSensor
color_image
depth_image_32f
depth_image_16u
label_image
X_WB


CHEEZIT
Domino
SUGAR

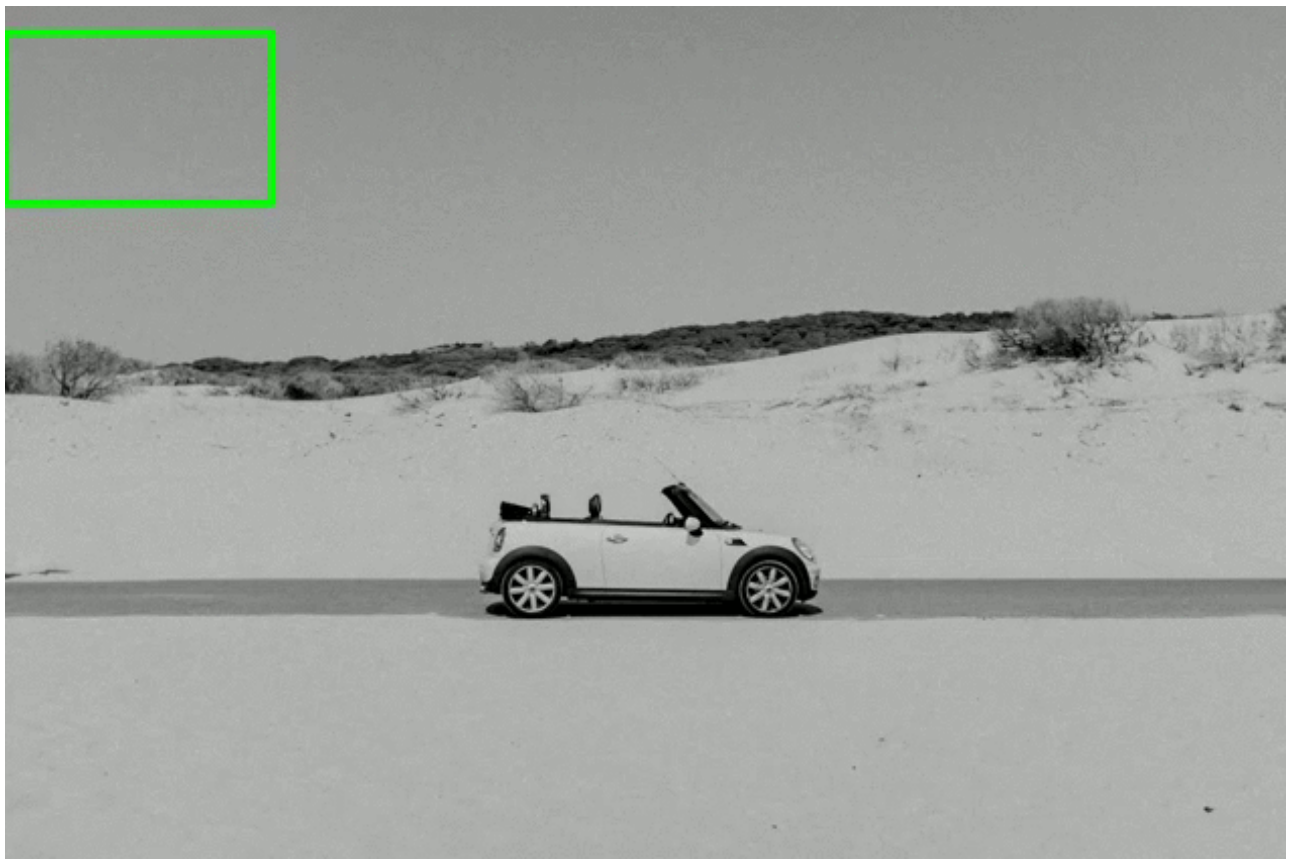




(a) Training envs with randomized visual rendering. (b) Test time.

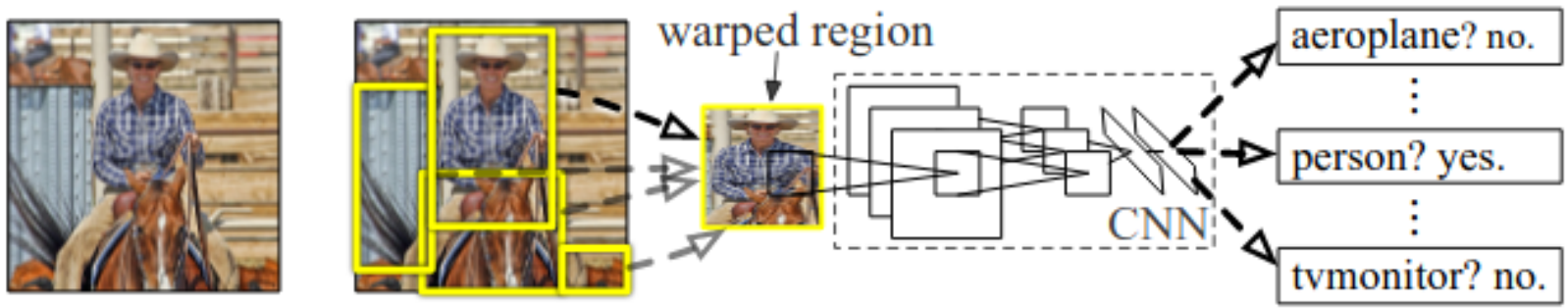
image from https://arxiv.org/abs/2012.02055

Object classification ⇒ detection (sliding window)



source: https://towardsdatascience.com/understanding-regions-with-cnn-features-r-cnn-ec69c15f8ea7

R-CNN: *Regions with CNN features*



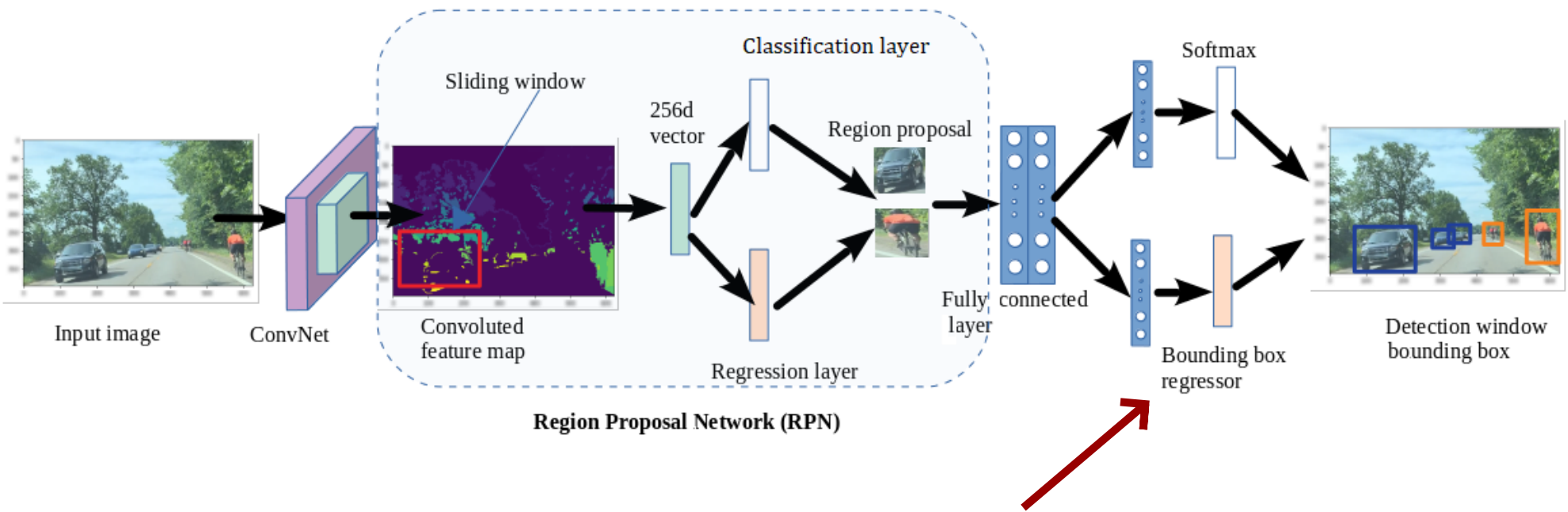
1. Input image

2. Extract region proposals (~2k)

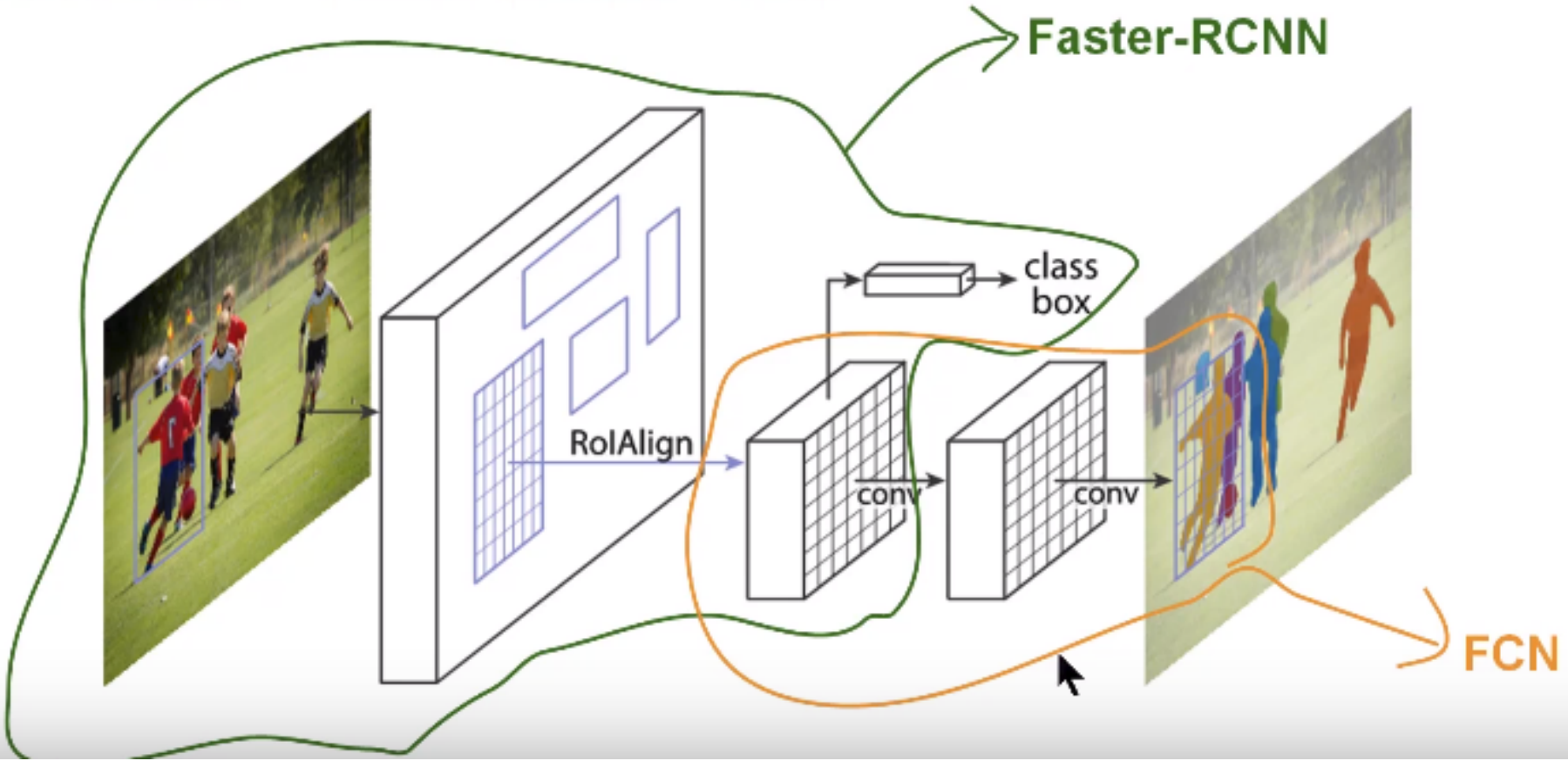
3. Compute CNN features

4. Classify regions

Faster R-CNN adds a "region proposal network"

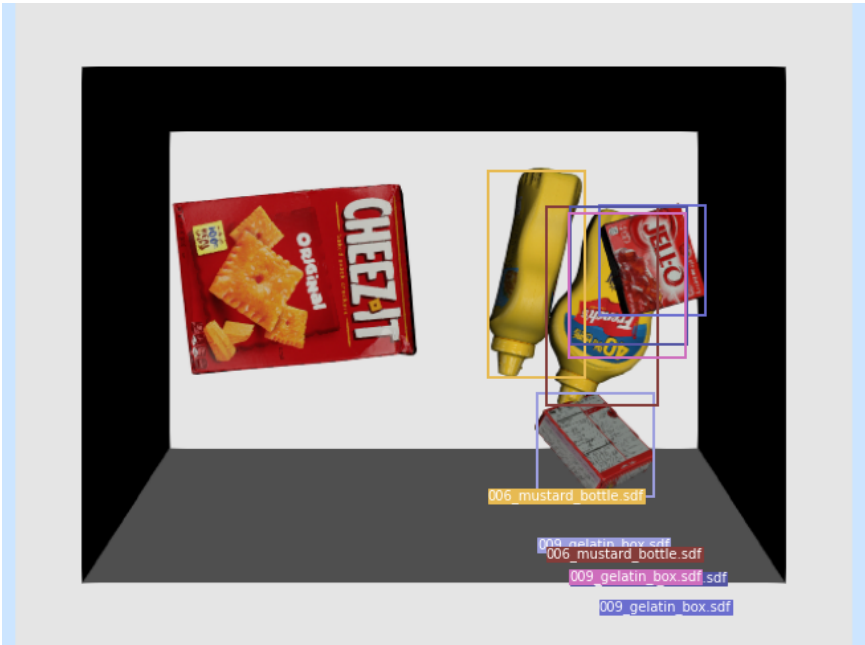


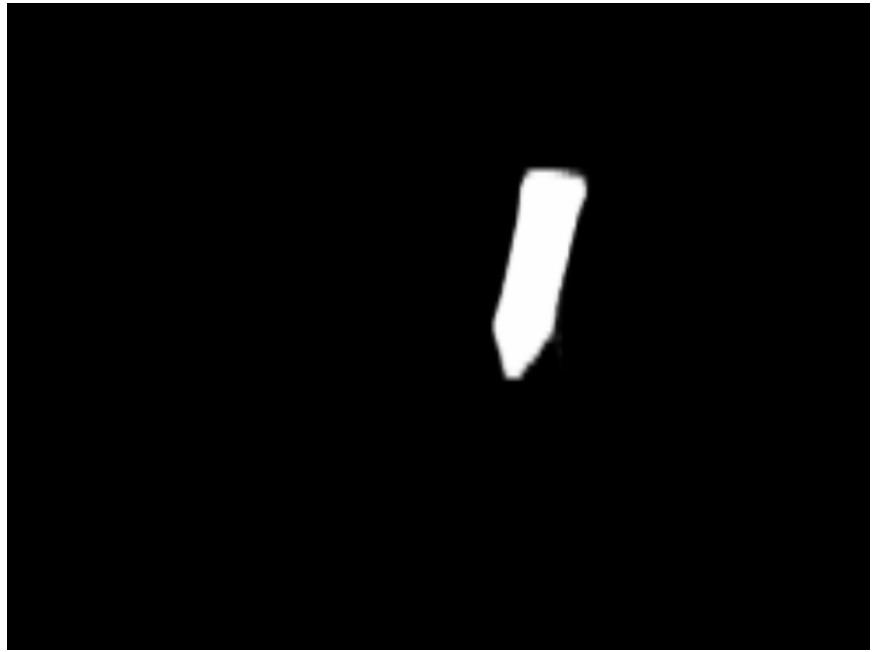
Mask R-CNN → Faster R-CNN + FCN



source: https://www.analyticsvidhya.com/blog/2018/07/building-mask-r-cnn-model-detecting-damage-cars-python/

Pick up the mustard bottles...

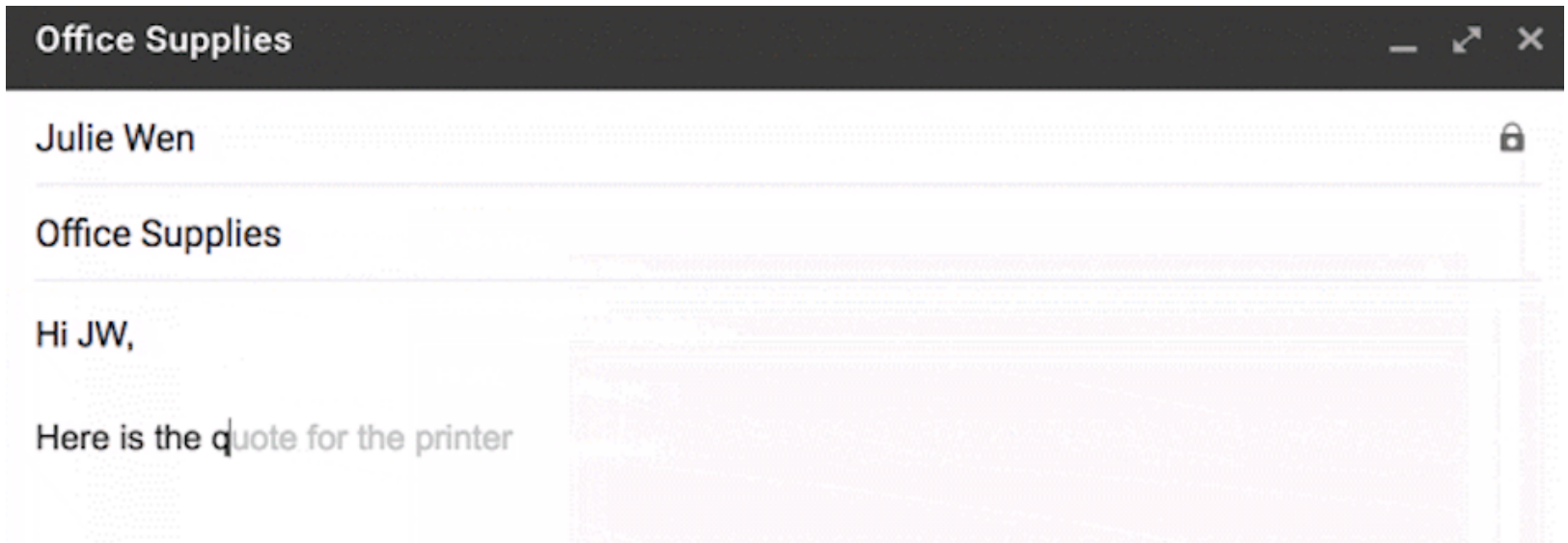




1. Segmentation + ICP => model-based grasp selection
2. Segmentation => antipodal grasp selection

"Self-supervised" learning

Example: Text completion



No extra "labeling" of the data required!

GPT-4 is "just" doing next-word prediction

Next-token-prediction

The model is given a sequence of words with the goal of predicting the next word.

Example:
Hannah is a ____

Hannah is a *sister*
Hannah is a *friend*
Hannah is a *marketer*
Hannah is a *comedian*

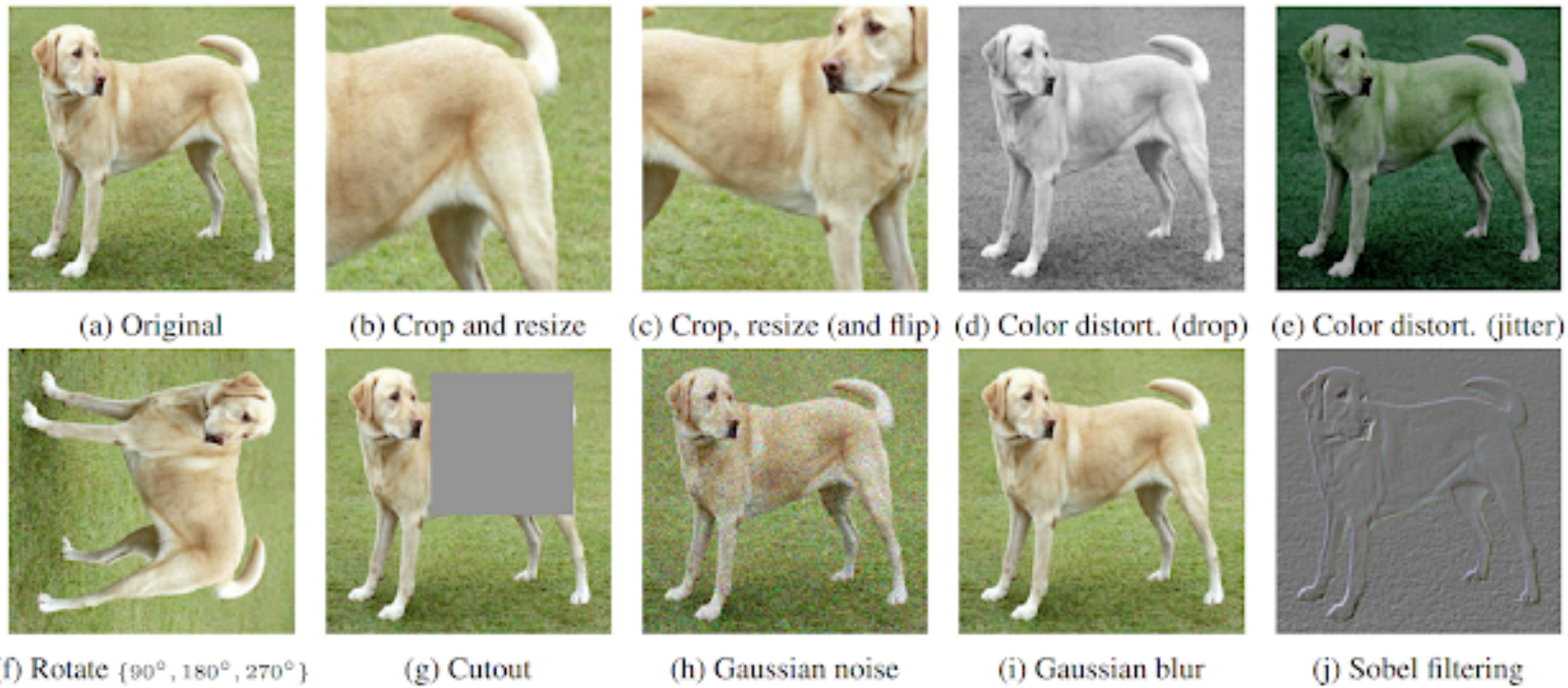
Masked-language-modeling

The model is given a sequence of words with the goal of predicting a 'masked' word in the middle.

Example
Jacob [mask] reading

Jacob *fears* reading
Jacob *loves* reading
Jacob *enjoys* reading
Jacob *hates* reading

Example: SimCLR



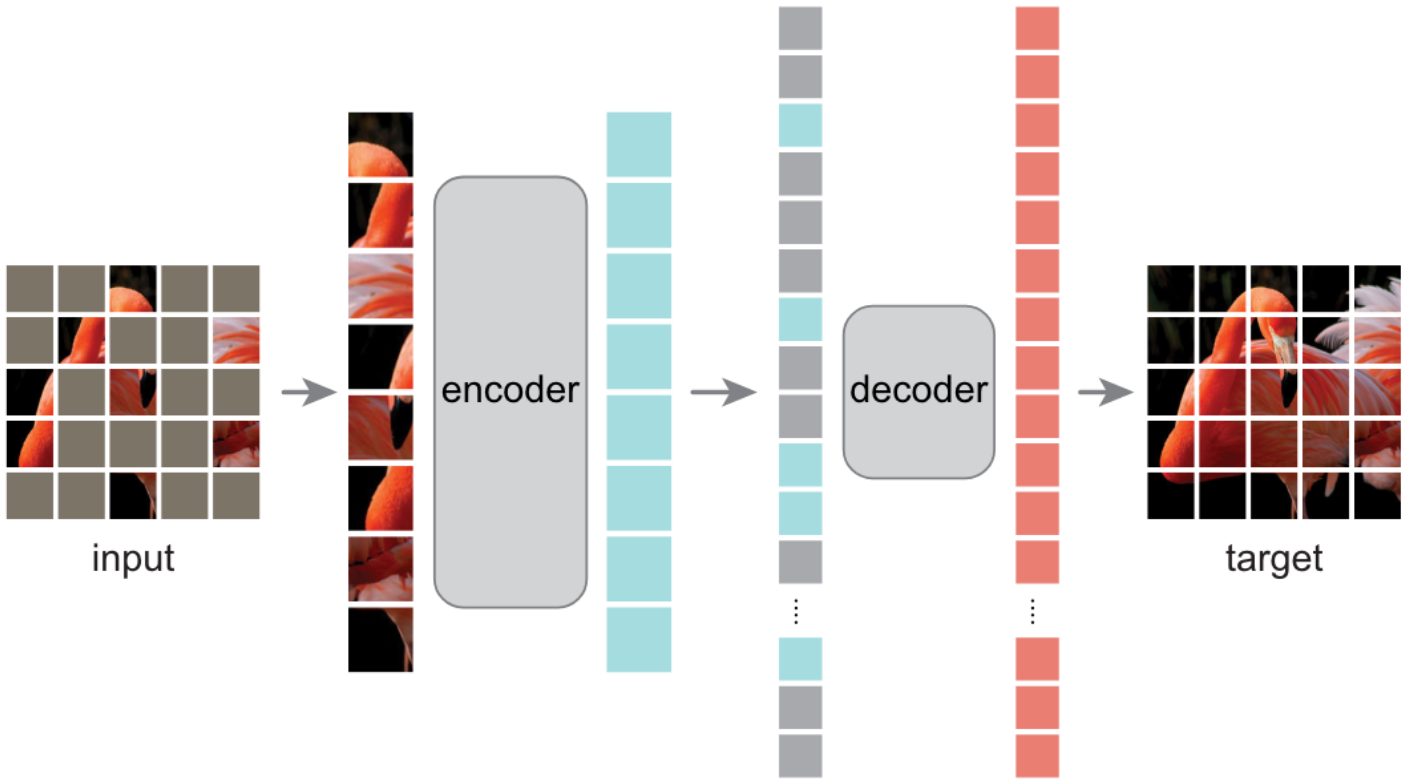
https://ai.googleblog.com/2020/04/advancing-self-supervised-and-semi.html

Example: SimCLR

"Contrastive visual representation learning"

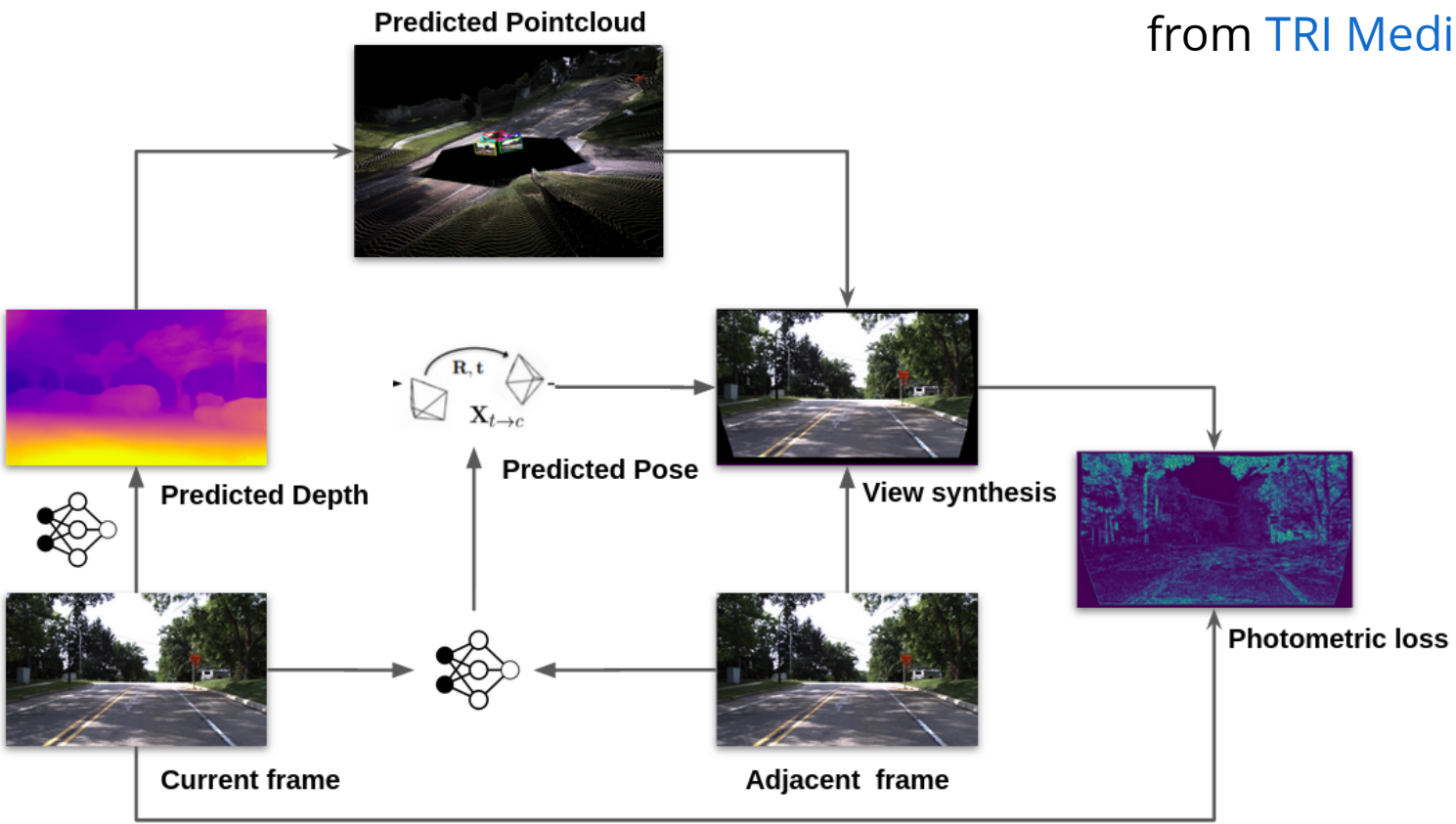
https://ai.googleblog.com/2020/04/advancing-self-supervised-and-semi.html

Masked Auto-Encoders (MAE)



Example: Monocular Depth Estimation

from TRI Medium Blog Post



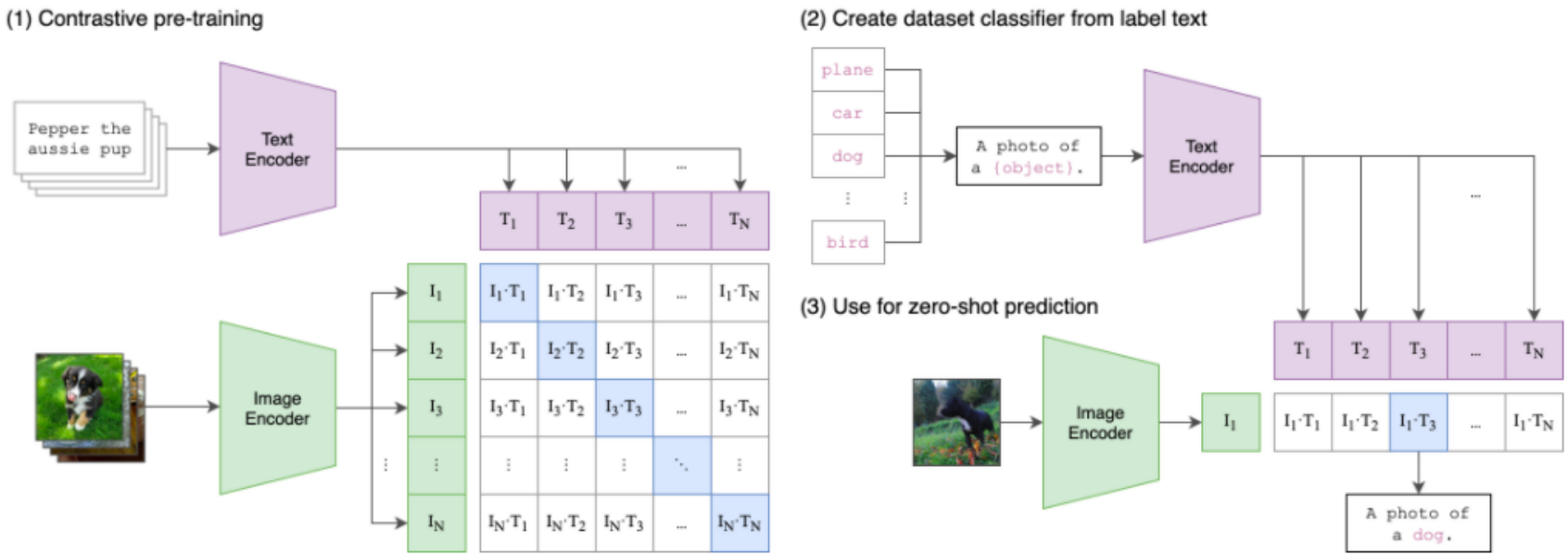
Foundation models

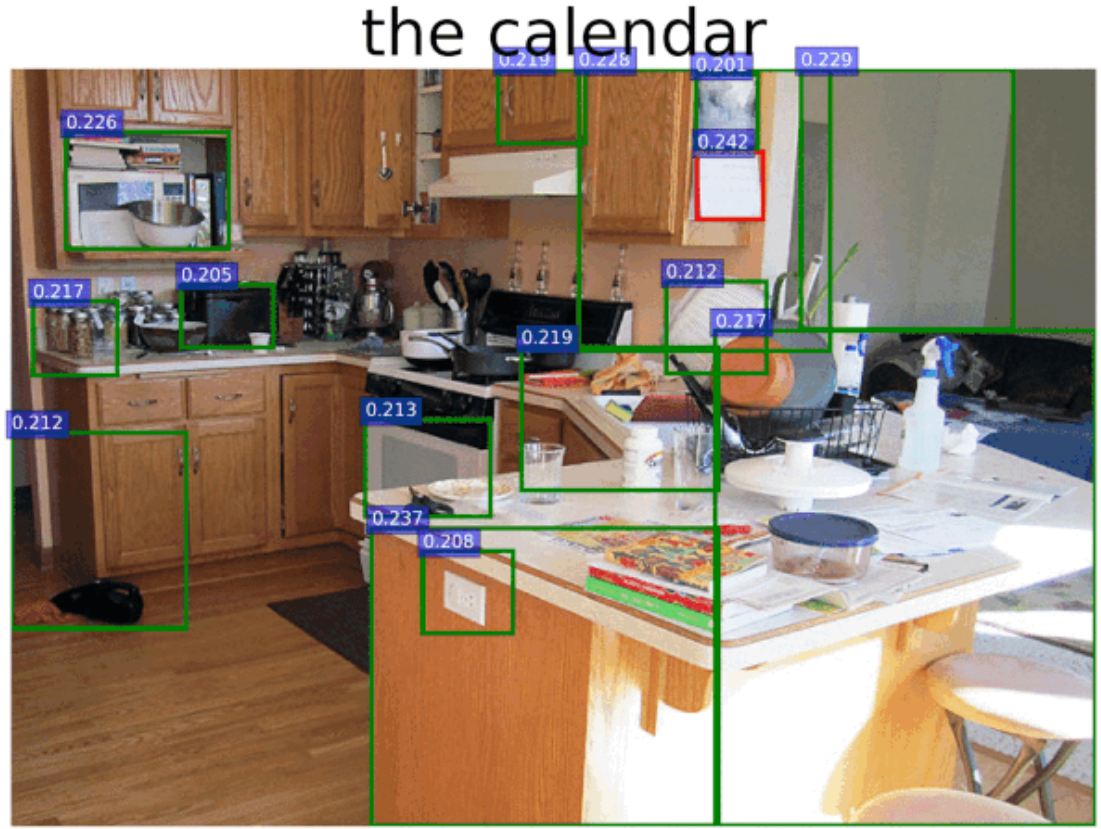
CLIP

[Blog] [Paper] [Model Card] [Colab]

CLIP (Contrastive Language-Image Pre-Training) is a neural network trained on a variety of (image, text) pairs. It can be instructed in natural language to predict the most relevant text snippet, given an image, without directly optimizing for the task, similarly to the zero-shot capabilities of GPT-2 and 3. We found CLIP matches the performance of the original ResNet50 on ImageNet "zero-shot" without using any of the original 1.28M labeled examples, overcoming several major challenges in computer vision.

Approach





quick experiments using CLIP "out of the box" by Kevin Zakka

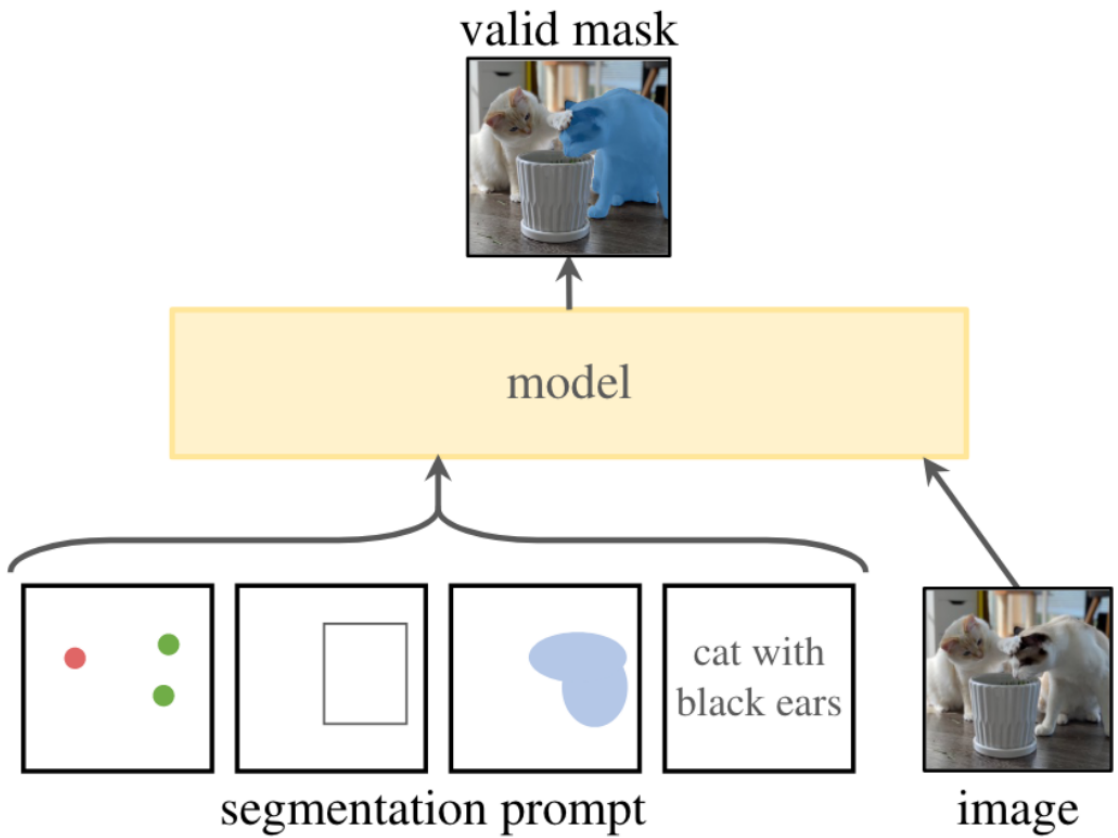
GPT-4V(ision)



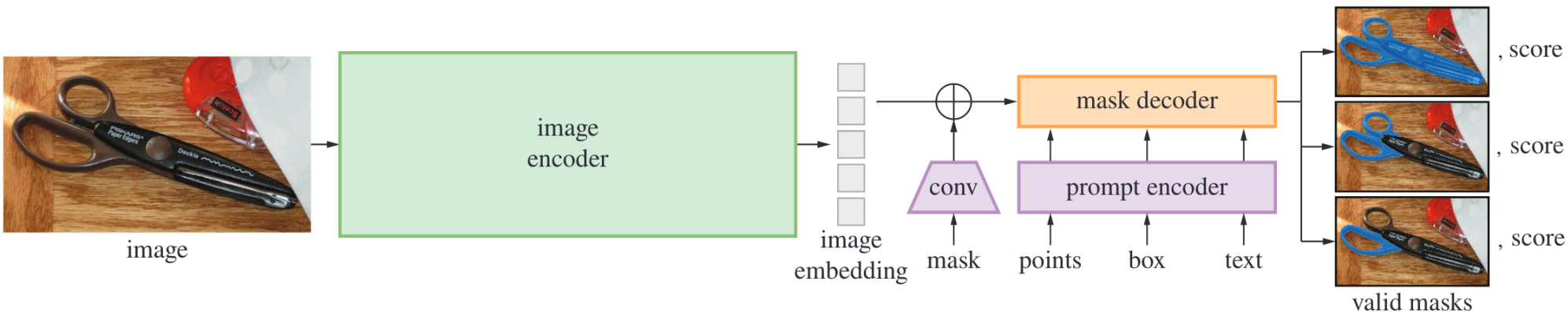


I need to clean the whiteboard. Can you give me step-by-step instructions using these objects?

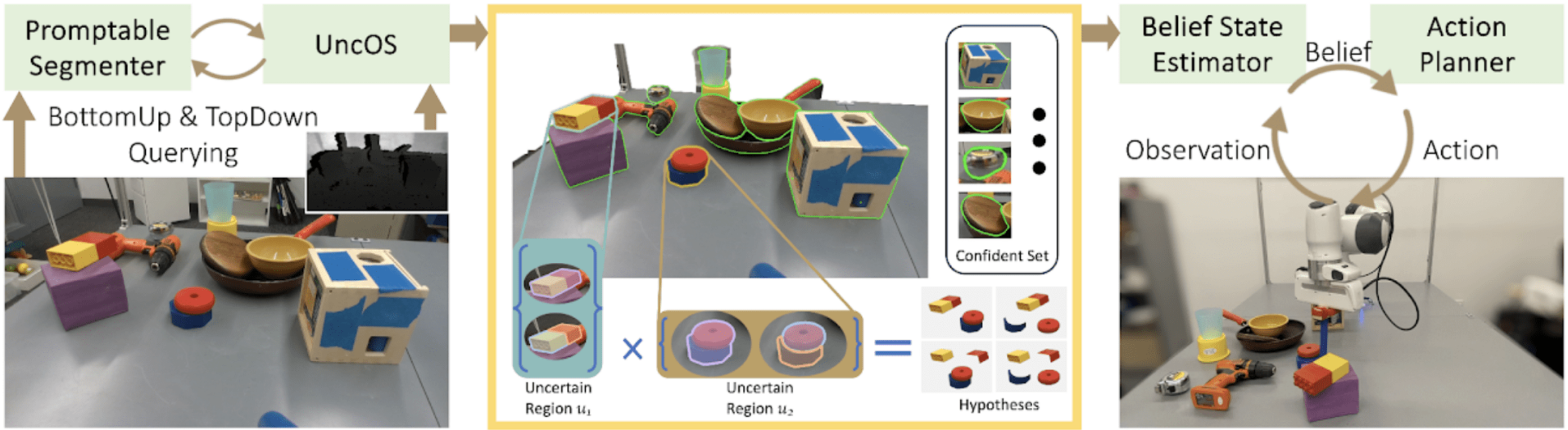
Segment Anything (Meta)



Segment Anything



Open-source release doesn't accept text. You need a wrapper… e.g. Grounded Segment Anything



UncOS

Fang et al, IROS 2024

https://www.youtube.com/embed/2I3zSwjeNOw?enablejsapi=1

https://www.youtube.com/embed/y6VYa87HKQY?enablejsapi=1

https://www.youtube.com/embed/Bp4ha76H8ME?enablejsapi=1

SceneComplete: Open-World 3D Scene Completion in Complex Real-World Environments for Robot Manipulation

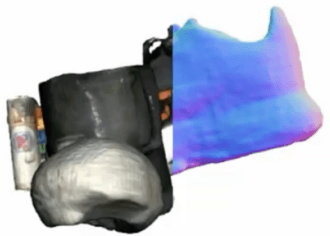
Aditya Agarwal[1], Gaurav Singh[2], Bipasha Sen[1],
Tomás Lozano-Pérez[1], Leslie Pack Kaelbling[1]

[1]MIT CSAIL, [2]IIIT-Hyderabad

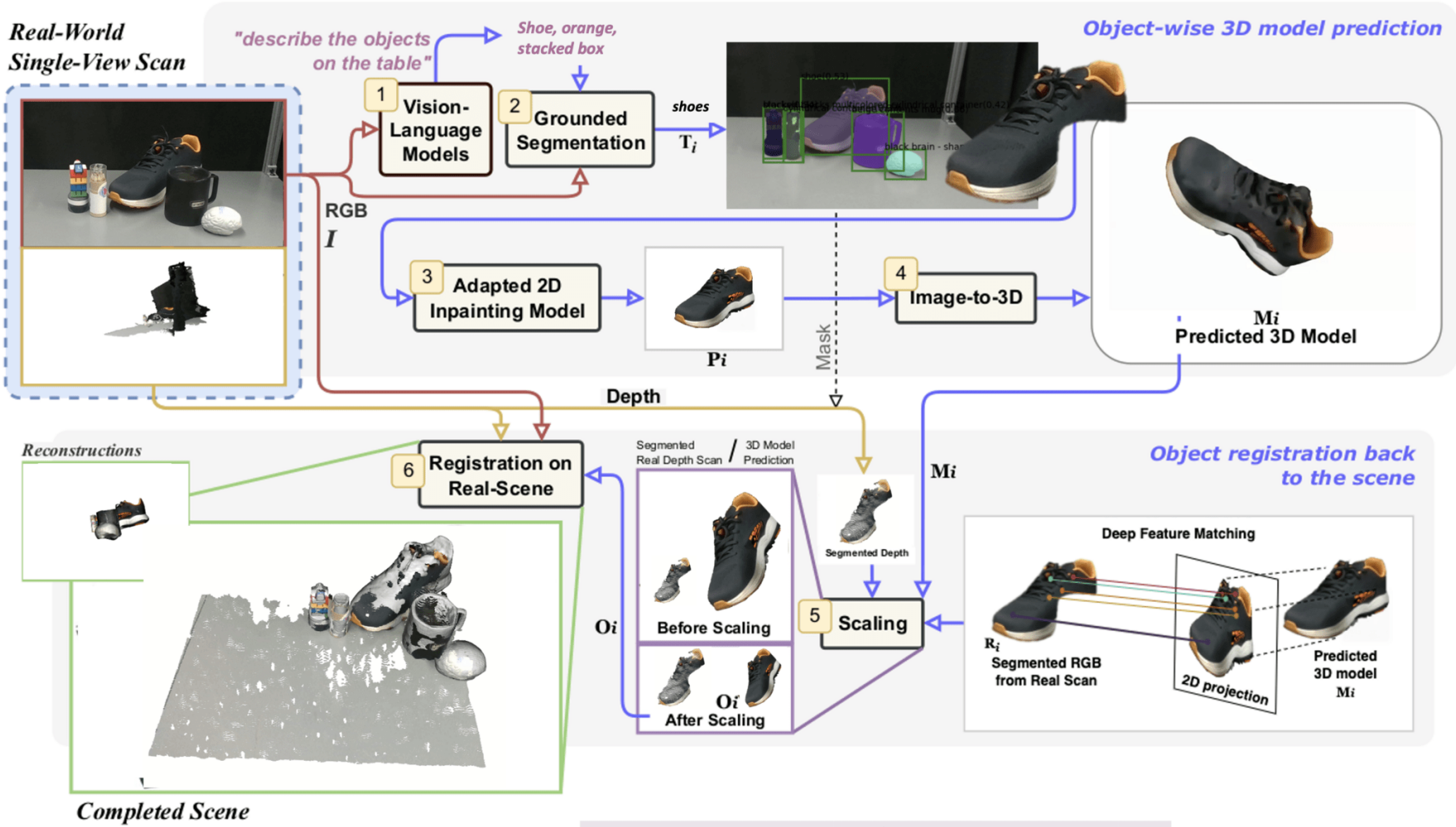








Object-based scene understanding: no prior object models



Aditya Agarwal, et al, SceneComplete, RAL